Q1. JEE Main 2026 (21 January Shift 2)

The r.m.s. speed of oxygen molecules at 47°C is equal to that of the hydrogen molecules kept at ____ °C. (Mass of oxygen molecule/mass of hydrogen molecule $= 32/2$)

(1) -235 (2) -253 (3) -100 (4) -20

Q2. JEE Main 2026 (28 January Shift 2)

The mean free path of a molecule of diameter 5×10^{-10} m at the temperature 41°C and pressure 1.38×10^5 Pa, is given as ____ m. (Given $k_B = 1.38 \times 10^{-23}$ J/K).

(1) $10\sqrt{2} \times 10^{-8}$ (2) $2\sqrt{2} \times 10^{-8}$
(3) $2\sqrt{2} \times 10^{-10}$ (4) 2×10^{-8}

Q3. JEE Main 2026 (22 January Shift 2)

Consider two boxes containing ideal gases A and B such that their temperatures, pressures and number densities are same. The molecular size of A is half of that of B and mass of molecule A is four times that of B. If the collision frequency in gas B is 32×10^{18}/s then collision frequency in gas A is ____ /s.

(1) 2×10^8 (2) 8×10^8
(3) 4×10^8 (4) 32×10^8

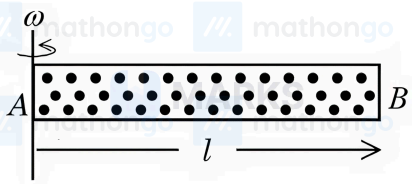
Q4. JEE Main 2026 (22 January Shift 1)

The volume of an ideal gas increases 8 times and temperature becomes $(1/4)^{\text{th}}$ of initial temperature during a reversible change. If there is no exchange of heat in this process $(\Delta Q = 0)$ then identify the gas from the following options (Assuming the gases given in the options are ideal gases):

(1) O_2 (2) NH_3 (3) CO_2 (4) He

Q5. JEE Main 2026 (22 January Shift 1)

A cylindrical tube AB of length l, closed at both ends contains an ideal gas of 1 mol having molecular weight M. The tube is rotated in a horizontal plane with constant angular velocity ω about an axis perpendicular to AB and passing through the edge at end A, as shown in the figure. If P_A and P_B are the pressures at A and B respectively, then (Consider the temperature is same at all points in the tube)



(1) $P_B = P_A$
(2) $P_B = P_A \exp\left(M\omega^2 l^2/3RT\right)$
(3) $P_B = P_A \exp\left(M\omega^2 l^2/RT\right)$
(4) $P_B = P_A \exp\left(M\omega^2 l^2/2RT\right)$

Q6. JEE Main 2026 (24 January Shift 1)

A gas of certain mass filled in a closed cylinder at a pressure of 3.23 kPa has temperature 50°C. The gas is now heated to double its temperature. The modified pressure is ____ Pa. Note: Volume is constant.

ANSWERS AND SOLUTIONS

1. (2) **2.** (2) **3.** (3) **4.** (4) **5.** (4) **6.** 3730

1. **(2)** The root-mean-square speed of gas molecules is: $v_{rms} = \sqrt{\frac{3RT}{M}}$, where T is absolute temperature and M is molar mass.

For oxygen at $T_1 = 47^\circ C = 320$ K with $M_{O_2} = 32$ g/mol:

$v_{rms,O_2} = \sqrt{\frac{3R(320)}{32}}$

For hydrogen with $M_{H_2} = 2$ g/mol at temperature T_2:

$v_{rms,H_2} = \sqrt{\frac{3RT_2}{2}}$

Equating the RMS speeds:

$\sqrt{\frac{3R(320)}{32}} = \sqrt{\frac{3RT_2}{2}}$

$\frac{320}{32} = \frac{T_2}{2}$

$T_2 = 2 \times 10 = 20$ K

Converting to Celsius: $t_2 = 20 - 273 = -253^\circ C$, which is option (2).

2. **(2)** Mean free path is calculated using $\lambda = \frac{k_B T}{\sqrt{2}\pi d^2 P}$

Given: $d = 5 \times 10^{-10}$ m, $T = 41^\circ C = 314$ K, $P = 1.38 \times 10^5$ Pa, $k_B = 1.38 \times 10^{-23}$ J/K

$\lambda = \frac{1.38\times10^{-23}\times314}{\sqrt{2}\times\pi\times(5\times10^{-10})^2\times1.38\times10^5}$

$= \frac{1.38\times10^{-23}\times314}{\sqrt{2}\times\pi\times25\times10^{-20}\times1.38\times10^5}$

Numerator: 433.32×10^{-23}

Denominator: $1.414 \times 3.14159 \times 34.5 \times 10^{-15} = 153.18 \times 10^{-15}$

$\lambda = \frac{433.32\times10^{-23}}{153.18\times10^{-15}} = 2.83 \times 10^{-8}$ m = $2\sqrt{2} \times 10^{-8}$ m

3. **(3)** Collision frequency $(z) = \sqrt{2}\pi d^2 N \sqrt{\frac{8RT}{\pi M}}$

Temp, N are same

$Z \propto \frac{d^2}{\sqrt{M}}$

$d_A = \frac{d_B}{2}$

$M_A = 4M_B$

$$\frac{Z_A}{Z_B} = \frac{d_A^2}{d_B^2} \times \frac{\sqrt{M_B}}{\sqrt{M_A}} = \left(\sqrt{\frac{M_B}{M_A}}\right)\left(\frac{d_A}{d_B}\right)^2$$
$$= \left(\sqrt{\frac{1}{4}}\right)\left(\frac{1}{2}\right)^2$$
$$\frac{Z_A}{Z_B} = \frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$$
$$\Rightarrow Z_A = \frac{32 \times 10^8}{8} = 4 \times 10^8/s$$

4. **(4)** For an adiabatic process ($\Delta Q = 0$) with an ideal gas: $TV^{\gamma-1} = \text{constant}$

Given: $V_f = 8V_i$ and $T_f = \frac{1}{4}T_i$

Applying the adiabatic relation:

$T_iV_i^{\gamma-1} = T_fV_f^{\gamma-1}$

$T_iV_i^{\gamma-1} = \frac{T_i}{4}(8V_i)^{\gamma-1}$

$1 = \frac{1}{4} \cdot 8^{\gamma-1}$

$4 = 8^{\gamma-1}$

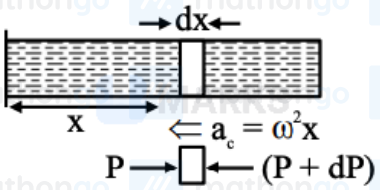
$2^2 = 2^{3(\gamma-1)}$

$2 = 3(\gamma - 1)$

$\gamma = \frac{5}{3} = 1.67$

This value corresponds to a monatomic gas. Among the options, only He is monatomic ($\gamma = 5/3$ for monatomic ideal gases). O_2 and N_2 are diatomic ($\gamma = 7/5$), CO_2 is polyatomic ($\gamma \approx 1.30$), and NH_3 is polyatomic.

5.



(4)

$A[(P + dP) - P] = (dm)(\omega^2 x)$

$dP = \frac{(dm)}{A}\omega^2 x$

$dP = \frac{(\rho)(A)(dx)\omega^2 x}{A}$

also $[PM = \rho RT]$

$\rho = \frac{PM}{RT}$

$dP = \left(\frac{PM}{RT}\right)\omega^2 x\, dx$

$\int_{P_A}^{P_B} \frac{dP}{P} = \frac{\omega^2 M}{RT}\int_0^{\ell} x\, dx$

$\ln\left(\frac{P_B}{P_A}\right) = \frac{\omega^2 \ell^2 M}{2RT}$

$P_B = P_A e^{\frac{M\omega^2\ell^2}{2RT}}$

6. **(3730)** Since its given that volume is constant, $P \propto T$.

$T_i = 50^\circ C = 323$ K

$T_f = 2 \times 50 = 100^\circ C = 373$ K

$\frac{P_f}{P_i} = \frac{T_f}{T_i}$

$\frac{P_f}{3.23 \text{ kPa}} = \frac{373}{323}$

$P_f = 3.23 \times \frac{373}{323} = 3.73 \text{ kPa} = 3730 \text{ Pa}$